

1. Problem 1 (6.2)

a.

$$89.5 \cdot \frac{1}{0.0440} = 2034 \text{ mol/m}^3$$

$$V_m = \frac{1}{2034} = 4.910 \cdot 10^{-4} \text{ m}^3/\text{mol}$$

$$b = 4.29 \cdot 10^{-5}$$

$$a = 3.66 \cdot 10^{-6}$$

$$P = \frac{RT}{V_m - b} - \frac{a}{V_m^2}$$

$$P = \frac{8.314 \cdot 10^{-5} \cdot 600}{4.910 \cdot 10^{-4} - 4.29 \cdot 10^{-5}} - \frac{3.66 \cdot 10^{-6}}{(4.910 \cdot 10^{-4})^2}$$

$$\boxed{P = 96.1 \text{ bar}}$$

b.

$$P = \frac{RT}{V_m}$$

$$P = \frac{8.314 \cdot 10^{-5} \cdot 600}{4.910 \cdot 10^{-4}}$$

$$\boxed{P = 101.5 \text{ bar}}$$

Ideal gas pressure is higher than the Van der Waals equation.

2. Problem 2 (6.4)

a.

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m}}$$

$$m = 32.07 + 6 \cdot 19 = 146.05 \text{ g/mol} = 146.05 \cdot \frac{1}{6.022 \cdot 10^{23}} \cdot \frac{1}{1000} = 2.43 \cdot 10^{-25}$$

$$\bar{v} = \sqrt{\frac{8k_B \cdot 400}{\pi \cdot 2.43 \cdot 10^{-25}}}$$

$$\boxed{\bar{v} = 241 \text{ m/s}}$$

b.

$$m = 0.028 \text{ kg/mol} = 4.65 \cdot 10^{-26}$$

$$\bar{v} = \sqrt{\frac{8k_B \cdot 400}{\pi \cdot 4.65 \cdot 10^{-26}}}$$

$$\boxed{\bar{v} = 550 \text{ m/s}}$$

c.

$$550 = \sqrt{\frac{8k_B \cdot T}{\pi \cdot 2.43 \cdot 10^{-25}}}$$

$$\boxed{T = 2086 \text{ K}}$$

3. Problem 3 (6.6)

a.

Given the side length = 20Å, distance required before n particles hits the same wall again = 40Å.

$$\begin{aligned}\frac{1}{2}mv^2 &= \frac{3}{2}k_B T \\ v^2 &= \frac{3k_B T}{m} \\ v_x^2 &= \frac{3k_B T}{m} \cdot \frac{1}{3} \\ v_x &= \sqrt{\frac{k_B T}{m}} \\ \frac{d}{t} &= \sqrt{\frac{k_B T}{m}} \\ \frac{1}{t} &= \sqrt{\frac{k_B T}{m}} \cdot \frac{1}{d} \\ \frac{1}{t} &= \sqrt{\frac{k_B \cdot 300}{\frac{0.028}{6.022 \cdot 10^{23}}}} \cdot \frac{1}{40 \cdot 10^{-10}} \\ \boxed{\frac{1}{t} &= 7.46 \cdot 10^{10} \text{ collisions/s}}\end{aligned}$$

b.

momentum change = $2mv_x$

$$\begin{aligned}2mv_x &= 2 \cdot \frac{0.028}{6.022 \cdot 10^{23}} \cdot \sqrt{\frac{k_B \cdot 300}{\frac{0.028}{6.022 \cdot 10^{23}}}} \\ \boxed{2mv_x &= 2.78 \cdot 10^{-23} \text{ kg m/s}}\end{aligned}$$

c.

$$\begin{aligned}P &= \frac{2mv_x}{d^2 t} \\ P &= \frac{2.78 \cdot 10^{-23}}{(20 \cdot 10^{-10})^2 \cdot \frac{1}{7.46 \cdot 10^{10}}} \\ \boxed{P &= 5.19 \cdot 10^5 \text{ Pa}}\end{aligned}$$

d.

$$\begin{aligned}P &= \frac{nk_B T}{V} \\ P &= \frac{1 \cdot k_B \cdot 300}{(20 \cdot 10^{-10})^3} \\ \boxed{P &= 5.19 \cdot 10^5 \text{ Pa}}\end{aligned}$$

4. Problem 4 (6.8)

a.

$$\begin{aligned}
 f(v) &= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp \left(\frac{-mv^2}{2k_B T} \right) \\
 \frac{df(v)}{dv} &= 0 \\
 \frac{df(v)}{dv} &= 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} \left(v^2 \cdot \frac{-2mv}{2k_B T} \cdot \exp \left(\frac{-mv^2}{2k_B T} \right) + 2v \cdot \exp \left(\frac{-mv^2}{2k_B T} \right) \right) \\
 0 &= v^2 \cdot \frac{-mv}{k_B T} + 2v \\
 \frac{mv^3}{k_B T} &= 2v \\
 mv^2 &= 2k_B T \\
 v &= \sqrt{\frac{2k_B T}{m}}
 \end{aligned}$$

b.

$$\begin{aligned}
 v &= \sqrt{\frac{2k_B \cdot 693}{\frac{0.032}{6.022 \cdot 10^{23}}}} \\
 v &= 600 \text{ m/s}
 \end{aligned}$$

5. Problem 5 (6.9)

a.

$$E = 600 \cdot 1.6 \cdot 10^{-19} = 9.6 \cdot 10^{-17} \text{ J}$$

$$v = \sqrt{\frac{2 \cdot 9.6 \cdot 10^{-17}}{\frac{0.028}{6.022 \cdot 10^{23}}}}$$

$$\boxed{v = 6.42 \cdot 10^4 \text{ m/s}}$$

b.

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m}}$$

$$T = \frac{\pi m}{8k_B} \cdot \bar{v}^2$$

$$T = \frac{\pi \cdot \frac{0.028}{6.022 \cdot 10^{23}}}{8k_B} \cdot (6.42 \cdot 10^4)^2$$

$$\boxed{T = 5.46 \cdot 10^6 \text{ K}}$$

6. Problem 6 (6.12)

$$F = \frac{P}{\sqrt{2\pi m k_B T}}$$

$$F = \frac{10^{-8} \cdot 10^5}{\sqrt{2\pi \cdot \frac{0.028}{6.022 \cdot 10^{23}} k_B \cdot 300}}$$

$F = 2.8 \cdot 10^{19} \text{ m}^{-2} \text{ s}^{-1}$

7. Problem 7 (6.14)

a.

$$Z = \sqrt{2}\sigma \sqrt{\frac{8k_B T}{\pi m}} \frac{P}{k_B T}$$

$$Z = \sqrt{2} \cdot 0.93 \cdot 10^{-18} \cdot \sqrt{\frac{8k_B \cdot 300}{\pi \cdot \frac{2 \cdot 0.0355}{6.022 \cdot 10^{23}}}} \cdot \frac{0.1 \cdot 10^5}{k_B \cdot 300}$$

$$Z = 9.5 \cdot 10^8 \text{ molecule}^{-1} \text{ s}^{-1}$$

$$\text{collisions} = 9.5 \cdot 10^8 \cdot 20$$

$$\boxed{\text{collisions} = 1.9 \cdot 10^{10} \text{ molecule}^{-1}}$$

b.

$$\bar{v} = \sqrt{\frac{8k_B \cdot 300}{\pi \cdot \frac{2 \cdot 0.0355}{6.022 \cdot 10^{23}}}}$$

$$\boxed{\bar{v} = 299 \text{ m/s}}$$

c.

$$\lambda = \frac{k_B T}{\sqrt{2}\sigma P}$$

$$\lambda = \frac{k_B \cdot 300}{\sqrt{2} \cdot 0.93 \cdot 10^{-18} \cdot 0.1 \cdot 10^5}$$

$$\boxed{\lambda = 3.15 \cdot 10^{-7} \text{ m}}$$

$$\lambda = \frac{\bar{v}}{Z}$$

$$\lambda = \frac{299}{9.5 \cdot 10^8}$$

$$\boxed{\lambda = 3.15 \cdot 10^{-7} \text{ m}}$$

8. Problem 8 (6.19)

$$\begin{aligned}
 n &= \frac{PV}{k_B T} \\
 -\frac{dn}{dt} &= \frac{-V}{k_B T} \frac{dP}{dt} = FA = \frac{PA}{\sqrt{2\pi m k_B T}} \\
 \frac{dP}{dt} &= \frac{-PA k_B T}{V \sqrt{2\pi m k_B T}} \\
 \int_{P_0}^P \frac{dP}{P} &= \int_0^t -\frac{A}{V} \sqrt{\frac{k_B T}{2\pi m}} dt \\
 \ln\left(\frac{P}{P_0}\right) &= -\frac{A}{V} \sqrt{\frac{k_B T}{2\pi m}} t \\
 A &= \frac{1}{100^2} = 0.001 \\
 V &= \frac{4}{3} \cdot \pi \cdot 0.1^3 = 0.00419 \\
 \ln\left(\frac{10^{-4}}{10}\right) &= -\frac{0.001}{0.00419} \sqrt{\frac{k_B \cdot 300}{2\pi \frac{0.018}{6.022 \cdot 10^{23}}}} t \\
 \boxed{t = 3.25 \text{ s}}
 \end{aligned}$$